



SSO Integration for ANNA DARPAN

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Coforge	Integration With SSO		Integration Details Document
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1 Introduction

In Anna Darpan, the Single Sign-On (SSO) process will be implemented & managed by Keycloak, an open-source identity and access management solution, designed to enable seamless and secure user authentication across different FCI internal applications.

Anna Darpan leverages LDAP as the central user identity repository, integrated with Keycloak, to provide a unified authentication layer across all integrated applications.

Anna Darpan serves as the centralized authentication platform, implemented using Keycloak, allowing users to log in once and access various FCI internal applications, such as HRMS, without needing to log in again for each one.

2 Scope:

The integration of external applications with Anna Darpan for user authentication, the technical approach leverages a **simplified authorisation code grant flow**. This process involves a series of redirects and exchanges between the external application, the user's browser, and Anna Darpan (Keycloak).

Additionally, LDAP acts as the backbone, ensuring user identity consistency and centralized management. User credentials are created once in LDAP and reused across all connected applications.

i. Integration with Applications:

Client applications rely on Anna Darpan for centralized authentication. Users log in once through Anna Darpan SSO, and the same session is reused across all connected applications, eliminating the need for multiple logins. Each application validates user identity using tokens issued by Keycloak.

ii. Technical Approach:

In this integration, each application is registered as a client in Keycloak, which enables Anna Darpan to manage authentication centrally. The Authorization Code Grant flow (OIDC) is used, where users authenticate through Anna Darpan and applications receive an authorization code that is exchanged for tokens.

The SSO integration enables Anna Darpan to serve as the central authentication hub, allowing users to log in once and access multiple applications seamlessly. This unified approach improves transparency, reduces the need for repeated logins, and ensures secure, real-time session management, thereby empowering stakeholders with a simplified and efficient user experience.

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3 Prerequisites for SSO Integration

System Integrators (SIs) and stakeholders must meet the following requirements to integrate with Anna Darpan SSO:

i. Login ID Specification:

- a. The **HRMS Employee ID** will serve as the login ID for all Anna Darpan SSO authentications.
- b. All applications that need to integrate with Anna Darpan SSO must maintain the HRMS Employee ID for each Food Corporation of India (FCI) employee in their databases to map user identities to Anna Darpan.
- c. The HRMS Employee ID must be synchronized with the LDAP repository managed by Anna Darpan for consistency.

ii. Password Management:

- a. SSO passwords are managed exclusively by Anna Darpan via Keycloak and LDAP.
- b. Integrated applications must not store or manage passwords locally. Passwords are maintained centrally in LDAP for security and consistency.

iii. Basic Requirements for Stakeholders:

- a. **Keycloak Client Registration:** Register the application as a client in Keycloak with a unique client ID, client secret, and redirect URI(s). Contact the Anna Darpan team for registration support.
- b. **OIDC Authorization Code Flow:** Implement the OIDC Authorization Code Flow to redirect users to Anna Darpan for authentication and handle the callback with the authorization code.
- c. **User Synchronization:** Ensure user data, including HRMS Employee IDs, is available in the Anna Darpan central LDAP repository. Coordinate with the Anna Darpan team to create credentials.
- d. **Token Validation:** Validate access tokens from Keycloak using the token introspection endpoint or JSON Web Key Set (JWKS) before granting access.
- e. **Redirect URI Configuration:** Provide a secure callback URL (e.g., <https://<application>/callback>) for Anna Darpan redirects.
- f. **Secure Communication:** Use HTTPS for all API calls to Anna Darpan and Keycloak to ensure secure data transmission.

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- g. **Session Management:** Implement session handling to maintain user sessions after token validation for seamless access.

4 End-to-End Flow (Step-by-Step)

- i. The user tries to log in to another FCI internal application that is integrated with Anna Darpan SSO.
- ii. The application shall redirect unauthenticated users to the Anna Darpan Single Sign-On (SSO) URL, appending the client_id & redirect_uri as a query parameter for authentication.
- iii. Upon entering their credentials, the user is authenticated by Keycloak. If the verification is successful, Keycloak redirects the user back to the application, providing an authorization code. (Credentials are verified against LDAP through Keycloak).
- iv. The application will exchange the authorization code and its client credentials with Keycloak, which validates the request and returns an access token along with a refresh token.
- v. The application backend validates the tokens, establishes a session, and grants the user secure access without requiring further logins.

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5 Flow Diagram

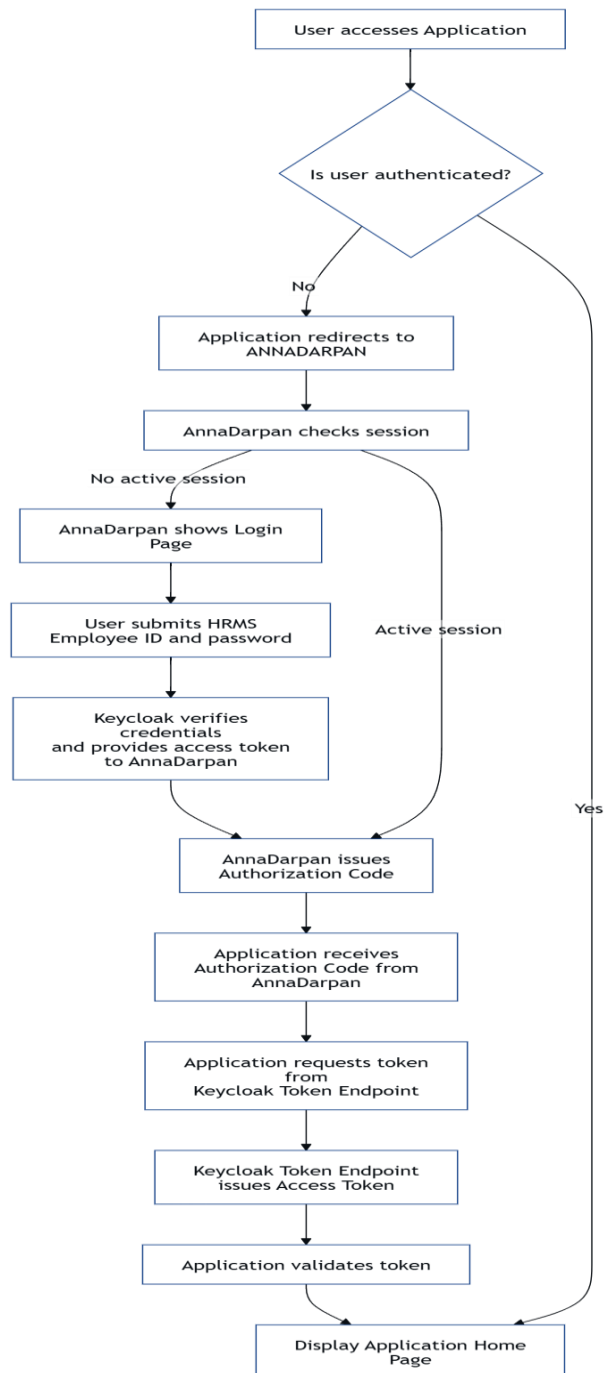


Figure 1: Direct Access to Application

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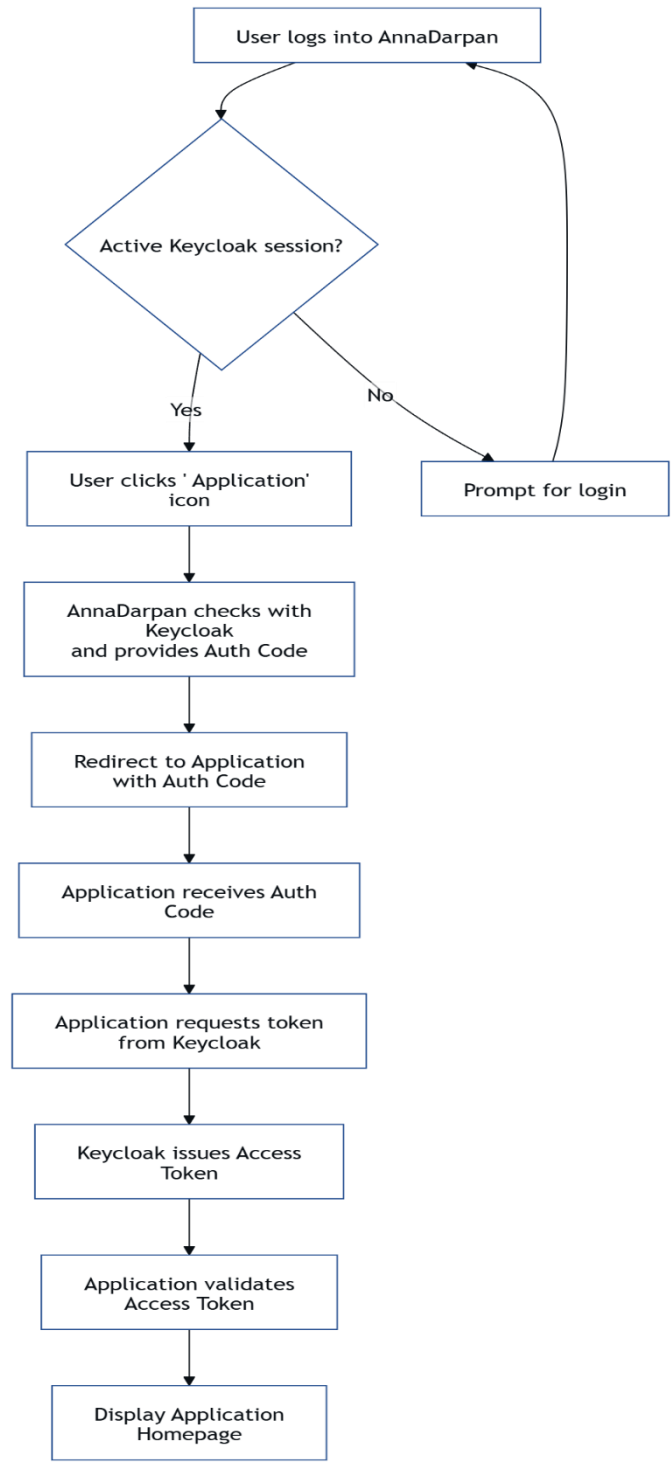


Figure 2: Access Application from AnnaDarpan

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6 External Application setup for SSO Integration

6.1 External Application Initiates Authentication with ANNADARPAN

The external application starts the SSO process by redirecting the user to AnnaDarpan's Keycloak Authorization Endpoint using the OIDC Authorization Code Flow.

Note: During registration of new applications in Keycloak, users of that application are required to be synced with Anna Darpan LDAP using the common attribute, i.e., HRMS Employee ID. While authentication will be handled by Keycloak, each application needs to manage its own authorization logic based on the user roles and permissions. Applications should map user roles and permissions based on their internal role management systems.

- i. The external application constructs a URL pointing to Anna Darpan's authorization endpoint with the required parameters.
- ii. The user's browser is redirected to this URL. If the user is not already authenticated, Anna Darpan prompts them with a login page.
- i.
- ii. **Parameters:**
 - a. **client_id:** Unique identifier of the application registered in Keycloak (e.g., app-client).
 - b. **redirect_uri:** Callback URL for post-authentication redirect (e.g., <https://<application>/callback>).
 - c. **response_type:** Must be set to **code** for the Authorization Code Flow.
 - d. **scope:** Includes openid (required), with optional scopes like profile or email.
 - e. **state:** Optional parameter to maintain state (e.g., xyz123).
- iii. **Example URL:**

https://annadarpan.in/authorize?client_id=app-client&redirect_uri=https://<application>/callback&response_type=code&scope=openid%20profile&state=xyz123

The external application redirects the user's browser to this URL. If the user is not authenticated, ANNA DARPAN displays a login page. After successful authentication, ANNA DARPAN processes the request and redirects back.

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6.2 AnnaDarpan Redirects to the External Application

After successful authentication, ANNA DARPAN (Keycloak) redirects the user's browser back to the external application's redirect_uri with an authorization code and other parameters.

- i. ANNA DARPAN authenticates the user (via login or an existing session).
- ii. Keycloak generates an authorization code and redirects the user to the external application's callback URL.

Redirect URL Example:

https://<application>/callback?authCode=<authorization_code>&username=<hrms_employee_id>&clientId=app-client&state=xyz123

Parameters (appended as query parameters):

- i. **code:** Short-lived authorization code for token exchange
- ii. **username:** The authenticated user's HRMS Employee ID
- iii. **clientId:** The application's client ID
- iv. **state:** The state value from the initial request, if provided(optional)

ANNA DARPAN redirects the user's browser to this URL. The external application's frontend receives these parameters and forwards them to its backend for token exchange

6.3 External Application Exchanges Authorization Code for Tokens

The application's frontend receives the authorization code and forwards it to its backend, which exchanges the code for an access token by calling Keycloak's api.

- i. The frontend sends the authorization code and other parameters to the backend.
- ii. The backend makes a secure POST request to Keycloak's token endpoint to exchange the code for an access token.

6.3.1 Handling Authorization Code in the External Application

- i. The frontend extracts the code, username, clientId, and state from the redirect URL.
- ii. It sends a POST request to the external application's backend API with these parameters.

```
{
  "authCode": "<authCode>",
  "username": "<username>",
  "clientId": "<clientId>"
}
```

6.3.2 Backend to Keycloak Custom Token Endpoint

- i. The frontend extracts the code, username, clientId, and state from the redirect URL.

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- ii. It sends a POST request to the external application's backend API with these parameters.

cURL Example:

```
curl --location
'https://devkeycloak.annadarpan.in/realms/AnnaDarpan/AnnaDarpan/getAccessToken' \
--header 'Content-Type: application/json' \
--data '{
  "username": "<username>",
  "clientId": "<clientId>",
  "authCode": "<authCode>",
  "secret": "<clientSecret>"
}'
```

Example Curl (Please replace with actual values):

```
curl --location
'https://devkeycloak.annadarpan.in/realms/AnnaDarpan/AnnaDarpan/getAccessToken' \
--header 'Content-Type: application/json' \
--data '{
  "username":"100005",
  "clientId":"ad-hrms",
  "authCode":"6GMWSBDL4NFCA3EODXMCK4YEEQOYQR7MDO7LSAI=",
  "secret":"ccMY5aoUb5Y5278A5ur1Qq2CAehr"
}'
```

7 SSO Integration Onboarding Guide

To onboard an application with AnnaDarpan SSO, follow these steps:

- i. **Register Application in Keycloak:** Define a unique client ID, redirect URIs, and assign necessary roles. Contact the AnnaDarpan team for assistance.
- ii. **Ensure LDAP Connectivity:**

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- a. Verify that HRMS Employee IDs are synchronized with the central LDAP repository.
- iii. **Configure OIDC Authorization Code Flow:**
 - a. Redirect users to AnnaDarpan for authentication and handle the callback with the authorization code.
- iv. **Implement Token Validation:**
 - a. Validate tokens with Keycloak's introspection endpoint or JWKS before granting access.
- v. **Test and Deploy:**
 - a. Test integration in the AnnaDarpan development environment (<https://.annadarpan.in>).
 - b. Verify both direct access and access from the AnnaDarpan dashboard.

8 Open Questions

Not Applicable.